



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Techniques

Hive is an SQL Like query language that enables SQL queries on large volumes of **structured** data. It has 4 functions and features.

- Query
- Data summarization
- Analysis
- ETL (extract, transform and Load)



Hive also supports MapReduce to perform more sophisticated analysis, users can choose between **MapReduce** or **Tez** or **Spark** framework as their execution backend.

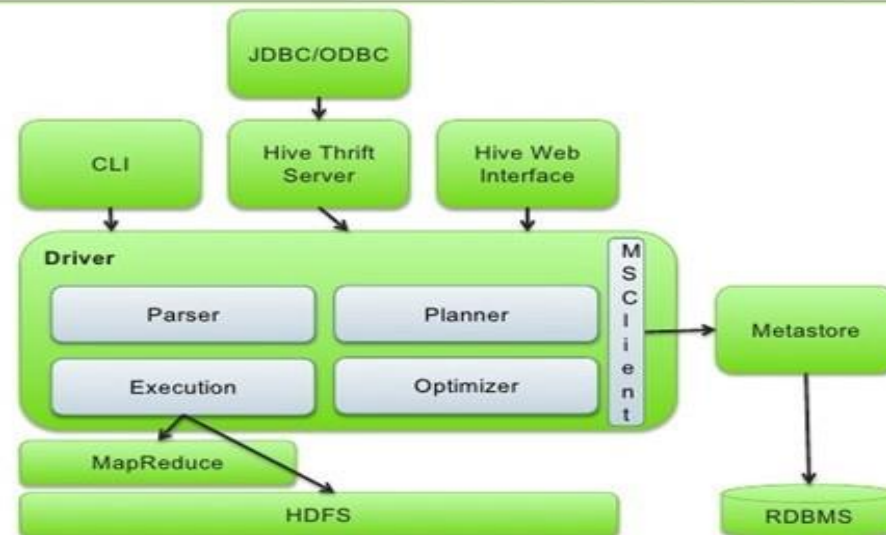
Hive is like tables in a relational database. Data in tables is serialized and each table as corresponding HDFS directory. Each table can be subdivided into partitions that determine how data is distributed within sub-directory of the table directory. Data within partition can be further broken down into buckets.

HIVE Architecture

4 main components

1. Hadoop core components
 - Hdfs, MapReduce
2. Metastore
3. Driver
4. Hive Clients

Apache Hive Architecture



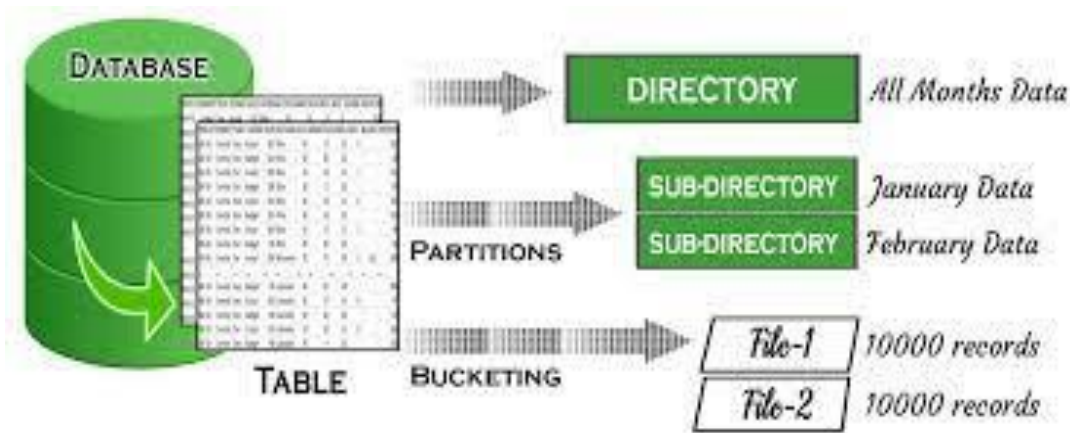
How Hive Works

1. **Query Compilation:** The query is parsed and transformed into an execution plan. This plan consists of multiple stages and tasks that need to be executed to process the query.
2. **Query Optimization:** The execution plan is optimized to improve performance. Hive performs various optimizations, such as predicate pushdown, join reordering, and column pruning, to minimize the amount of data read and processed.
3. **Query Execution:** The optimized execution plan is executed. Hive translates the tasks in the plan into a series of **MapReduce or Tez jobs**, depending on the execution engine being used.
4. **MapReduce or Tez Execution:** Hive leverages either MapReduce or Tez to process the data based on the execution plan. The data is read from the **HDFS**, and processed in **parallel across multiple nodes in the cluster**.
5. **Data Processing:** The MapReduce or Tez jobs process the data according to the query logic defined in the HQL. This involves tasks like filtering, aggregating, joining, and sorting the data.
6. **Result Output:** Once the data processing is complete, Hive generates the result set based on the query and writes it to the specified output location. This can be a file in HDFS, a table in a Hive metastore, or a user-defined destination.

Hive – HDFS – Data Model

The Hive Data Model contains following components

- Database
- Tables
- Partitions
 - Partition means dividing a table into parts based on the value of a partition column. This makes it faster to do queries on slice of data
- Buckets
 - Buckets give extra structure to the data that may be used for efficient queries.
 - A join of two tables that are bucketed on the same columns, including the join column can be implemented as a Map Side Join.



Hive – Core Components - MapReduce



When we run a query, it will run a Map Reduce job by converting or compiling the query into a java class file, building a jar file and execute this jar file.

Hive – Core Components - MetaStore



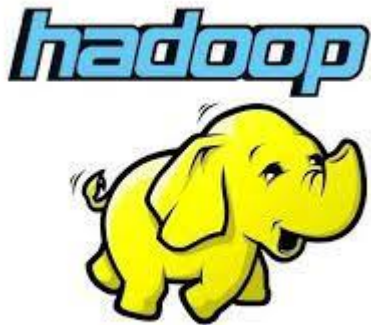
Metastore is a name space for tables. This is a crucial part of the hive as all the metadata information related to the hive such as details related to the table, columns, partition, location is present as part of it.

Usually , the Metastore is available as part of the relational database like mySQL

Hive is a query engine or abstraction of MapReduce, not a database and doesn't have its own storage to store data

Hive = HADOOP + SQL

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SQL

1 Cloudera.com : <https://www.cloudera.com/tutorials/getting-started-with-hdp-sandbox.html>

★ Please also refer to references slide at the end of this presentation for credits and citations for the contents that are not exactly copied but taken as a knowledge or reference

HIVE continued !

Apache **ORC** (Optimized **R**ow **C**olumnar) is a fast columnar storage file format for Hadoop Workloads

the file format provides a highly efficient way to store Hive data designed to improve performance when Hive is reading, writing and processing data.

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CREATE TABLE <table name> ... STORED AS ORC...

Following chain of events happens when you import a table using ORC file format.

1. a table created using ORC file format
2. A temporary table is created using TEXT FILE format to store data from CSV file
3. Data is copied from the temporary table to target ORC table
4. Temporary table is dropped

1 Cloudera.com : <https://www.cloudera.com/tutorials/getting-started-with-hdp-sandbox.html>



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HIVE

Learning resources



tutorialspoint.com/hive/hiveql_group_by.htm

tutorialspoint.com/sql/sql-sum-function.htm

tutorialspoint.com/apache_drill/aggregate_function_countdistinct.htm

tutorialspoint.com/hive/hiveql_select_where.htm

Apache Pig is a platform for analyzing large data sets that consists of a **high-level language** for expressing data analysis programs, coupled with **infrastructure** for evaluating these programs.

The salient property of Pig programs is that their structure is amenable to substantial parallelization, which in turns enables them to handle very large data sets.

At the present time, Pig's infrastructure layer consists of a compiler that produces sequences of Map-Reduce programs, for which large-scale parallel implementations already exist (e.g., the Hadoop subproject). Pig's language layer currently consists of a textual language called **Pig Latin**, which has the following key properties:

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- **Ease of programming.** It is trivial to achieve parallel execution of simple, "embarrassingly parallel" data analysis tasks. Complex tasks comprised of multiple interrelated data transformations are explicitly encoded as data flow sequences, making them easy to write, understand, and maintain.
- **Optimization opportunities.** The way in which tasks are encoded permits the system to optimize their execution automatically, allowing the user to focus on semantics rather than efficiency.
- **Extensibility.** Users can create their own functions to do special-purpose processing.



```
C: > GBC > courses > CSI > 202306 > material > Topic 2 -Apache Hadoop > pigcode.pig
1  -- Load data from a file
2  data = LOAD 'input_data.txt' USING PigStorage(',') AS (name:chararray, age:int);
3
4  -- Filter records where age is greater than 25
5  filtered_data = FILTER data BY age > 25;
6
7  -- Group data by name
8  grouped_data = GROUP filtered_data BY name;
9
10 -- Compute the average age for each group
11 result = FOREACH grouped_data GENERATE group AS name, AVG(filtered_data.age) AS average_age;
12
13 -- Store the result in a file
14 STORE result INTO 'output_data.txt' USING PigStorage(',');
```

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1 Pig.apache.org.: <https://pig.apache.org/>



Please also refer to references slide at the end of this presentation for credits and citations for the contents that are exactly copied or paraphrased for the topic and context

Pig- Use Cases

Pig Latin, the scripting language used in Apache Pig, is designed for data processing on large-scale datasets. It provides a flexible and expressive way to perform data transformations, filtering, aggregations, and analytics. Here are some common use cases where Pig Latin can be beneficial:

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Pig.apache.org.: <https://pig.apache.org/>



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Pig- Architecture

If we see the top-level overview of Pig, then Pig is an abstraction over MapReduce. Pig runs on Hadoop. So, it makes use of both the Hadoop Distributed File System (HDFS) and Hadoop's processing system, MapReduce.

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Apache Pig was designed by Yahoo as it is easy to learn and work with. So, Pig makes Hadoop quite easy. Apache Pig was developed because MapReduce programming was getting quite difficult, and many MapReduce users are not comfortable with declarative languages. Now, Pig is an open-source project under Apache.



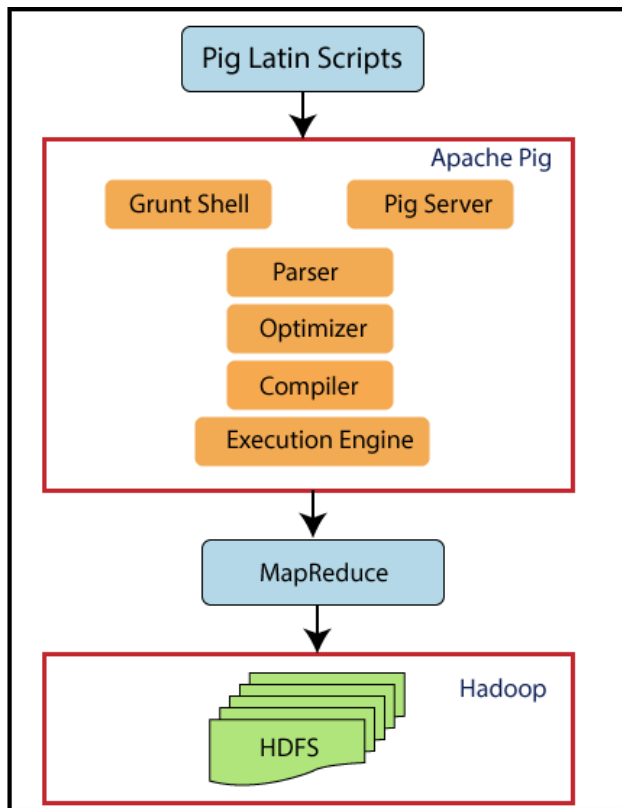
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Pig.apache.org.: <https://pig.apache.org/>



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Pig- Architecture



Pig sits on top of Hadoop. Pig scripts can be run either on Grunt shell or on the Pig server. Execution engine of Pig the passes optimizes and compiles the script and finally converts into MapReduce jobs. It uses HDFS to store intermediate data between MapReduce jobs and then writes its output to HDFS.

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Pig.apache.org.: <https://pig.apache.org/>



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Hive – Pig – Comparison – 1/2

Hive and Pig are both high-level data processing tools in the Apache Hadoop ecosystem, but they differ in their approach to data processing and the query languages they use

Query Language

1. Hive: Hive uses Hive Query Language (HQL), which is similar to SQL. It allows users to write queries using SQL-like syntax, making it familiar to those who are already comfortable with SQL.
2. Pig: Pig uses Pig Latin, a data flow scripting language. Pig Latin is a procedural language that focuses on the data flow and transformations rather than the traditional SQL-like declarative queries.

Data Processing Model

1. Hive: Hive is designed for structured data processing. It provides a relational database-like structure called tables, and users can define schemas and perform SQL-like operations such as filtering, aggregating, joining, and sorting on these tables.
2. Pig: Pig is designed for data processing on any kind of data, including structured, semi-structured, and unstructured data. It operates on data in the form of data flows, where users define a series of transformations to process the data. Pig's flexibility allows it to handle complex data processing tasks.

Hive – Pig – Comparison : 2/2

Data Optimization and Execution

1. Hive: Hive compiles queries into a series of MapReduce or Tez jobs. It optimizes queries using techniques like predicate pushdown, join reordering, and column pruning. With the introduction of Hive LLAP, it also provides in-memory caching and interactive query processing capabilities for faster execution.
2. Pig: Pig also compiles queries into MapReduce or Tez jobs. It optimizes the execution plan using a logical and physical plan optimization process. Pig leverages the Pig Latin language to optimize and execute complex data processing tasks efficiently.

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User Community and Ecosystem

1. Hive: Hive has a larger user community and a mature ecosystem due to its SQL-like interface and integration with existing data warehousing tools. It is widely used in organizations that are familiar with SQL and have existing SQL-based workflows.
2. Pig: Pig has a smaller user community compared to Hive, but it is popular among users who prefer a procedural programming approach for data processing tasks. Pig's flexibility makes it suitable for ad-hoc data processing and exploratory analysis.

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Pig.apache.org.: <https://pig.apache.org/>



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Sample Pig Code

-- Load data from a file

```
data = LOAD 'input_data.txt' USING PigStorage(',') AS (name:chararray, age:int);
```

-- Filter records where age is greater than 25

```
filtered_data = FILTER data BY age > 25;
```

-- Group data by name

```
grouped_data = GROUP filtered_data BY name;
```

-- Compute the average age for each group

```
result = FOREACH grouped_data GENERATE group AS name, AVG(filtered_data.age) AS average_age;
```

-- Store the result in a file

```
STORE result INTO 'output_data.txt' USING PigStorage(',');
```

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Pig.apache.org.: <https://pig.apache.org/>



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Furter reading on pig

<https://www.javatpoint.com/pig-data-types>

https://www.tutorialspoint.com/apache_pig/apache_pig_limit_operator.htm

https://www.tutorialspoint.com/apache_pig/apache_pig_order_by.htm

https://www.tutorialspoint.com/apache_pig/apache_pig_group_operator.htm

https://www.tutorialspoint.com/apache_pig/apache_pig_sum.htm

https://www.tutorialspoint.com/apache_pig/apache_pig_join_operator.htm

[Apache Pig - Union Operator - Tutorialspoint](#)

[Grouping Joining Apache pig - Beyond Corner](#)

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Pig.apache.org.: <https://pig.apache.org/>



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